

PS

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CSCI 4220 • MACHINE & DEEP LEARNING

Robby's Saturday

Week 1–3 • Six small problems, one long day

Due Wed 23 Sep, 11:59pm • 100 points • Submit a single PDF to Canvas

Three shall be the number thou shalt count, and the number of the counting shall be three. Four shalt thou not count, neither count thou two, excepting that thou then proceed to three. Five is right out.

— Monty Python and the Holy Grail, 1975

HOW THIS WORKS

Six scenes from one Saturday. In each one Robby runs into a problem that an algorithm from the last three weeks happens to solve, mostly because he refuses to solve anything the normal way.

Every problem starts with **one worked example** — the same kind of step the problem then asks you to do, done in full. Copy the shape, change the numbers. No calculators and no code: every answer is an integer or a simple fraction, and if yours isn't, check your arithmetic.

One problem per page. Show your work, keep it short — most parts want a line or two.

THE DAY

7:04 a.m.	Robby makes a list	supervised or not?	16
8:30 a.m.	Office hours, three weeks running	least squares	17
10:00 a.m.	The pool, and a bad learning rate	gradient descent	17
1:30 p.m.	A fruit with no sticker	k -nearest neighbours	17
4:00 p.m.	The inbox	naïve Bayes	17
9:45 p.m.	Two tables	k -means	16

Submit one PDF to Canvas. Typed or neatly handwritten and scanned are both fine — legibility is graded, handwriting is not. You may discuss the problems with classmates, but write your solutions alone. AI tools are permitted and encouraged; if you use one, say so briefly at the top and say what it did for you.

1 7:04 a.m. — Robby makes a list

SHORT ANSWER 16 pts

Robby wakes up at 7:04, four minutes before his alarm, which he takes personally. He lies there losing an argument with the ceiling, gives up, and writes a list of things he would like to know today on the back of a receipt for \$61 of protein powder. Two of them are for Dr. Dan, who asked on Thursday. Robby said he would have them by Friday.

For each item, say whether it is **supervised** or **unsupervised**, and if supervised, whether it is **regression** or **classification**.

 HERE'S ONE DONE FOR YOU

“Will it rain this afternoon?” — Robby has years of weather records, each day labelled *rained* or *didn't*. He wants to predict which of those two words applies to today.

Supervised (past days come with the right answer attached), **classification** (the answer is one of two categories, not a number). The label y is the *rained* / *didn't-rain* tag on each past day.

- (a) How many students will come to office hours on Monday, going by the counts from the last few weeks.
- (b) Whether the email that just landed is spam or a real student question.
- (c) Which of his 34 students naturally group together by project interest, so he can suggest teams. Nobody has decided the groups ahead of time.
- (d) How fast he will swim the 100 free next month, given his times so far.
- (e) For each one you called supervised, name the label y — one short phrase, like the worked example does.

PROBLEM 1 • YOUR WORK

2 8:30 a.m. — Office hours, three weeks running

BY HAND 17 pts

Robby has been counting office-hours attendance. Week 2 was not a triumph: one student came, and he was looking for the registrar. Robby explained gradient descent to him anyway. Across the hall Dr. Sanders had a queue into the corridor, but Robby is choosing not to model that.

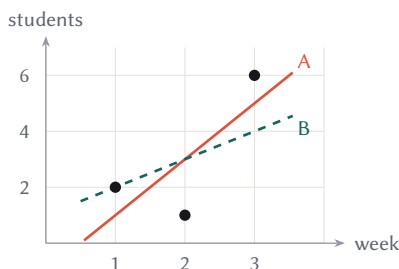
week x	1	2	3
students y	2	1	6

He draws two lines through it, likes both, and can defend either — which is the worst of the available outcomes:

Line A: $\hat{y} = -1 + 2x$

Line B: $\hat{y} = 1 + x$

The way to settle it is the *loss*: for each week take the residual $r_i = y_i - \hat{y}_i$, square it, and add the three squares up. Smaller loss means better fit.



Line A is steeper; line B is flatter and passes closer to the first two points. Which one wins depends on how badly the third point is missed — and squaring is what decides that.

HERE'S ONE DONE FOR YOU

A third line, $\hat{y} = 2 + x$, scored in full. Work across the table one week at a time — predict, subtract, square:

week x	actual y	predicted $\hat{y} = 2 + x$	residual $r = y - \hat{y}$	r^2
1	2	3	-1	1
2	1	4	-3	9
3	6	5	1	1
loss:				11

So this line scores 11. Now do the same for A and for B, and compare.

- Build that table for **line A** and again for **line B**, and report the loss of each. Which line fits better?
- Robby's colleague says that if you keep trying lines you will eventually find one with a loss of 0. Look at the picture and say in one sentence why that will never happen for this data.

PROBLEM 2 • YOUR WORK

3 10:00 a.m. — The pool, and a bad learning rate

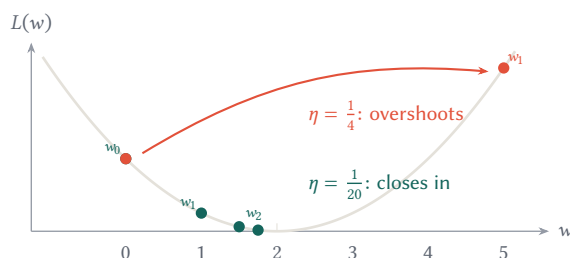
BY HAND 17 pts

Robby swims two sets: 1 lap in 2 minutes, 2 laps in 4 minutes. His pace is obviously 2 minutes per lap. A reasonable person would stop here. Robby is going to make a computer derive it from scratch instead, because it is Saturday, he brought his laptop to the pool deck, and nobody present has the authority to stop him.

The model is $\hat{y} = wx$, and for this data the loss and its slope are

$$L(w) = (w - 2)^2 + (2w - 4)^2 = 5(w - 2)^2, \quad L'(w) = 10(w - 2).$$

Gradient descent repeats $w \leftarrow w - \eta L'(w)$ from $w_0 = 0$. We already know it should end up at $w = 2$; the question is whether it gets there.



The same bowl-shaped loss, walked two ways. A small step always lands nearer the bottom; a big one sails clean over it and comes down higher than it started.

HERE'S ONE DONE FOR YOU

Take $\eta = \frac{1}{10}$ (not one of the ones you need). Substituting,

$$w \leftarrow w - \frac{1}{10} \cdot 10(w - 2) = w - (w - 2) = 2.$$

So from $w_0 = 0$ we get $w_1 = 2$, and then $w_2 = 2$, $w_3 = 2$ — this particular step size lands exactly on the answer in one jump and stays there. Notice the method: substitute η , simplify the update into the form $w \leftarrow (\text{something})w + (\text{something})$, then apply it repeatedly.

- Take $\eta = \frac{1}{20}$. Show the update simplifies to $w \leftarrow \frac{1}{2}w + 1$, then compute w_1 , w_2 and w_3 as exact fractions. Are they heading toward 2?
- Now take $\eta = \frac{1}{4}$. Show the update becomes $w \leftarrow -\frac{3}{2}w + 5$, then compute w_1 , w_2 and w_3 . In one sentence, what is going wrong?
- Same loss, same starting point, same code. In one sentence, what is the *only* thing that made (b) fail?

'TIS BUT A SCRATCH

Robby did not invent this to pad the homework. In CA.01 — the coding assignment that follows this pset — a learning rate of 0.1 trains perfectly well on standardised features and sends the parameters to 10^{273} on raw ones. Identical code, identical data, one difference: the size of the numbers going in.

The rule here is $\eta < \frac{1}{5}$, and in general $\eta < 2/L$. That is what people mean by “scale your features first” — it is arithmetic, not superstition.

PROBLEM 3 • YOUR WORK

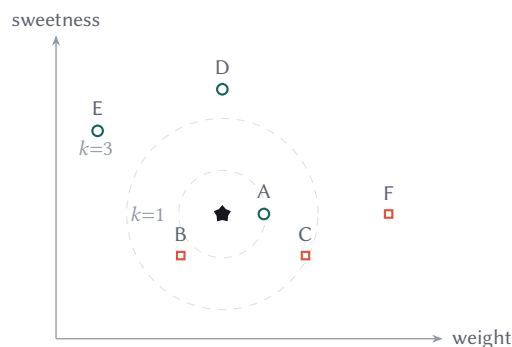
4 1:30 p.m. — A fruit with no sticker

BY HAND 17 pts

At the store Robby picks up a citrus that has lost its sticker. An employee is standing four feet away, visibly available, arguably employed for this exact purpose. Robby will not be consulting the employee. He weighs the fruit, consults the sweetness card, writes down (4, 3), and settles in. Six fruits in the bin still have their labels:

	A	B	C	D	E	F
(weight, sweetness)	(5, 3)	(3, 2)	(6, 2)	(4, 6)	(1, 5)	(8, 3)
label	lime	lemon	lemon	lime	lime	lemon

k -NN finds the k closest labelled examples and takes a majority vote.



Circles are lime, squares are lemon, the star is the mystery fruit. The inner ring holds its single nearest neighbour, the outer ring its nearest three — and they disagree.

HERE'S ONE DONE FOR YOU

Distance from the mystery fruit (4, 3) to A = (5, 3). Subtract coordinate by coordinate, square each, add:

$$(5 - 4)^2 + (3 - 3)^2 = 1^2 + 0^2 = 1.$$

So A sits at squared distance 1. Do the same for B: $(3 - 4)^2 + (2 - 3)^2 = 1 + 1 = 2$.

We can compare squared distances directly and never take a square root, because squaring keeps things in the same order — whatever is closest stays closest. It also keeps every number a whole number.

- Finish the job: squared distances from (4, 3) to C, D, E and F. List all six fruits nearest to furthest, then take a majority vote for $k = 1$ and $k = 3$. You will not get the same answer twice.
- Robby re-weighs everything on a scale that reads ten times finer, so every *weight* is multiplied by 10 while sweetness stays as it is. Recompute the squared distances and redo $k = 3$. What changed, and what should he have done to the data first?

In (b) you do not need to redo the subtraction. If a weight difference was -1 before, it is now -10 , so that part of the sum gets multiplied by 100. The sweetness part is untouched.

PROBLEM 4 • YOUR WORK

5 4:00 p.m. — The inbox

BY HAND 17 pts

Robby has labelled his last eight emails — four spam, four real — in pursuit of an ambition he has held since 2014: never opening his inbox again.

	# emails	contain “free”	contain “meeting”
spam	4	3	1
real	4	1	3

A new one arrives. It contains “free” and does *not* contain “meeting”. This is promising: every email Dr. Dan has ever sent contains the word “meeting”, so whatever this is, it is probably not another meeting.

Naïve Bayes scores each class by multiplying the prior by one factor per feature, then compares the two scores.

 HERE'S ONE DONE FOR YOU

Reading probabilities off the table. Of the 4 spam emails, 3 contained “free”, so

$$P(\text{free} \mid \text{spam}) = \frac{3}{4}.$$

Of the 4 spam emails, 1 contained “meeting”, so $P(\text{meeting} \mid \text{spam}) = \frac{1}{4}$ — which means

$$P(\text{no meeting} \mid \text{spam}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

And the prior is just how common spam is overall: $P(\text{spam}) = \frac{4}{8} = \frac{1}{2}$.

Building a score. Multiply the prior by one factor per feature, using the “no meeting” version because that is what the email actually looks like:

$$\text{spam score} = \underbrace{\frac{1}{2}}_{\text{prior}} \times \underbrace{\frac{3}{4}}_{\text{free}} \times \underbrace{\frac{3}{4}}_{\text{no meeting}} = \frac{9}{32}.$$

Now do the same for “real” and compare.

- Do the “real” half: read its three numbers off the table and build its score exactly the way the worked example built the spam one. Then divide each score by the sum of the two to get $P(\text{spam} \mid \text{this email})$.
- “Naïve” means the method assumes the two words show up independently of each other once you know the class. Give one reason to doubt that about real email.

PROBLEM 5 • YOUR WORK

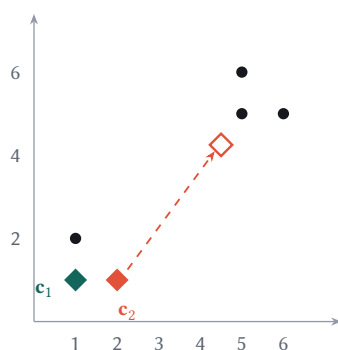
6 9:45 p.m. — Two tables

BY HAND 16 pts

It is quarter to ten on a Saturday night and Robby is making a seating chart. This was not the plan for the evening. Six students, two tables, and he wants similar interests sitting together. He has two survey scores for each:

$$(1, 1), (1, 2), (2, 1), (5, 5), (5, 6), (6, 5).$$

k -means with $k = 2$ repeats two steps: *put every student at the nearest centre, then move each centre to the average of its own students*. He starts both centres in the same corner. He knows this is a bad idea. He does it anyway, on the grounds that it is a quarter to ten on a Saturday: $c_1 = (1, 1)$ and $c_2 = (2, 1)$.



Both centres start jammed in the bottom-left corner. Watch what happens to c_2 in round 1 — it gets hauled out to the middle of nowhere before settling.

HERE'S ONE DONE FOR YOU

Where does student (5, 5) go? Squared distance to each centre:

$$\text{to } c_1 = (1, 1) : (5 - 1)^2 + (5 - 1)^2 = 16 + 16 = 32,$$

$$\text{to } c_2 = (2, 1) : (5 - 2)^2 + (5 - 1)^2 = 9 + 16 = 25.$$

$25 < 32$, so (5, 5) joins table C_2 .

And once everyone is assigned, a centre moves to the average of its own students — average the first coordinates, then average the second. If a table ended up with (2, 1) and (5, 5) on it, its new centre would be $\left(\frac{2+5}{2}, \frac{1+5}{2}\right) = \left(\frac{7}{2}, 3\right)$.

- Round 1.** Work out where the other five students go, then compute both new centres. Give them as exact fractions.
- Round 2.** Exactly one student changes tables. Say which one, give the two final centres, and explain how you know the algorithm is finished and Robby can go to bed. (Where one centre is obviously miles closer, just say so — you do not need every number.)

PROBLEM 6 • YOUR WORK